

THE MATHEMATICA CODE IBM.M

These notes explain the use of the Mathematica code `ibm.m`. It treats a generic model of N bosons of p different kinds with angular momenta $\ell_1, \dots, \ell_p$ and with interactions of arbitrary order k . With `ibm.m` it is possible to express symbolically or numerically matrix elements of a Hamiltonian in $\ell_1 \dots \ell_p$ -IBM as a linear combination of k -body interaction matrix elements. Also matrix elements of arbitrary tensor operators can be calculated.

The algorithms used in `ibm.m` are described in more detail in [1].

1. Necessary files

To run the `ibm.m` code the following files are needed:

- `racah.m`: Calculates Clebsch-Gordan coefficients, $3j$, $6j$, $9j$, $12j$ and $3nj$ symbols and some related angular-momentum objects. If the arguments are given as integers (for bosons) or half-odd-integers (for fermions), the result is returned as the square root of a rational number.
- `stats.m`: Calculates the number of states for N bosons of p different kinds or for N fermions distributed over p orbitals coupled to total angular momentum J .
- `cfpsen.m`: Calculates coefficients of fractional parentage (CFPs) in a seniority basis for a single kind of boson with integer angular momentum ℓ or for a single kind of fermion with half-odd-integer angular momentum j . The basic quantities are one-particle CFPs $(n-1) \times 1 \rightarrow n$, from which the more general $(n-k) \times k \rightarrow n$ CFPs are derived.
- `eigen.m`: Diagonalises a square symmetric matrix and returns its eigenvalues, ordered by increasing value, and the corresponding normalised eigenvectors.
- `ibm.m`: Constructs N -boson basis states in $\ell_1 \dots \ell_p$ -IBM and calculates matrix elements of a rotationally invariant, boson-number conserving Hamiltonian with k -body interactions and of tensor operators of arbitrary order that are not necessarily scalar nor boson-number conserving. The different possible forms of the Hamiltonian are detailed in these notes.

```
In[1]:= << Documents/mathematica/ibm_sdg.m
```

```
In[2]:= dimb[{0, 2, 4}, 10, 0]
```

```
Out[2]= 839
```

```
In[3]:= basb[{0, 2, 4}, 10, 0]
```

```
Out[3]=
```

```
{{{10, 0, 1, 0, 0}, {0, 0, 1, 0, 0}, {0, 0, 1, 0, 0}},  
 {{8, 0, 1, 0, 0}, {2, 0, 1, 0, 0}, {0, 0, 1, 0, 0}},  
 {{7, 1, 1, 0, 0}, {3, 3, 1, 0, 0}, {0, 0, 1, 0, 0}},  
 {{6, 0, 1, 0, 0}, {4, 0, 1, 0, 0}, {0, 0, 1, 0, 0}}, ... 832 ...,  
 {{0, 0, 1, 0, 0}, {0, 0, 1, 0, 0}, {10, 10, 3, 0, 0}},  
 {{0, 0, 1, 0, 0}, {0, 0, 1, 0, 0}, {10, 10, 4, 0, 0}},  
 {{0, 0, 1, 0, 0}, {0, 0, 1, 0, 0}, {10, 10, 5, 0, 0}}}
```

Size in memory: 0.5 MB

+ Show more

Show all

Iconize

Store full expression in notebook



```
In[4]:= Length[%]
```

```
Out[4]= 839
```

FIG. 1: With `dimb[{0,2,4},10,0]` one finds that the number of states in *sdg*-IBM for $N = 10$ bosons with angular momentum $J = 0$ is 839. With `basb[{0,2,4},10,0]` the corresponding basis states are constructed. The list is too long to be shown in its entirety but the length corresponds.

- `ibm_sd.m` & `ibm_sdg.m` & ...: Contains additional definitions that are specific to particular realisations of $\ell_1 \dots \ell_p$ -IBM. At the moment *sd*-IBM and *sdg*-IBM are implemented.
- `young.m`: Calculates group-theoretical properties such as Young tableaux manipulations, multiplications of representations, branching rules, plethysms, etc. These properties are only needed for comparison of the results of `ibm.m` with exactly solvable cases (see section 9.3).

To start a calculation, read the file for only one particular realisation, `ibm_sd.m` or `ibm_sdg.m` or ... From there all other necessary files are read. The current setting assumes that all files are in one folder called “Documents/mathematica”.

2. Construction of the basis

The number of states for N bosons with angular momentum J is calculated with the statement `dimb[{ $\ell_1, \dots, \ell_p$ },  $N, J$ ]`, where $\{\ell_1, \dots, \ell_p\}$ are the angular momenta of the bosons,

e.g. $\{0, 2\}$ for *sd*-IBM. If the set $\{\ell_1, \dots, \ell_p\}$ includes an *s* boson, it should come first, $\ell_1 = 0$.

Basis states are constructed with `basb` $[\{\ell_1, \dots, \ell_p\}, N, J]$, which generates an array of length `dimb` $[\{\ell_1, \dots, \ell_p\}, N, J]$. Each element of that array is itself an array of length p , the i th element of which refers to the sequence of quantum numbers $\{n_i, v_i, \alpha_i, J_i, K_i\}$ associated with the boson with angular momentum ℓ_i . This sequence consists of the boson number n_i , the seniority v_i , the multiplicity index α_i , the angular momentum J_i and the intermediate angular momentum K_i . For more details on the definition of basis states, see [1].

3. Definition of the interaction

The statement `genintb` $[\{\ell_1, \dots, \ell_p\}, k]$ generates a list of scalar interactions of order k . There are several options: `genintAb` generates all interactions while `genintHb` generates only those that are Hermitian. Similarly, `genintAPb` generates all parity-conserving interactions (see section 7 on how to impose conservation of parity) and `genintHPb` only those that are Hermitian. A default choice of interaction is made by `genintb`, which is currently set to `genintAb` and `genintAPb`.

With `genintb` the interactions are generated in the notation adopted in `basb`, which is rather cumbersome. The interactions can be generated in a simplified notation by `genintsb` with the same arguments as `genintb`. The same options as above are available, that is, `genintsAb`, `genintsHb`, `genintsAPb` and `genintsHPb`. The simplified notation is as follows:

- $k = 0$ generates a constant E0.
- $k = 1$ generates the single-boson energies in the notation $e[\ell]$, where ℓ is given in standard spectroscopic notation, that is, *s* for $\ell = 0$, *p* for $\ell = 1$, *d* for $\ell = 2$, etc.
- $k = 2$ generates all two-body interactions in the notation $v[\ell_{i_1}\ell_{i_2}, \ell_{i_3}\ell_{i_4}; L]$, where $\ell_{i_1}\ell_{i_2}$ and $\ell_{i_3}\ell_{i_4}$ specify the two bosons in bra and ket, respectively, and L is their coupled angular momentum. For all versions $\ell_1 \dots \ell_p$ -IBM these labels suffice to completely determine a two-boson state.

For the notation of matrix elements with $k > 2$, see section 10.1.

```
In[1]:= << Documents/mathematica/ibm_sd.m
```

```
In[6]:= genintHb[{0,2},3]
```

```
Out[6]:= {{{{3, 1, 1, 0, 0}, {0, 0, 1, 0, 0}}, {{3, 1, 1, 0, 0}, {0, 0, 1, 0, 0}}},
  {{{3, 1, 1, 0, 0}, {0, 0, 1, 0, 0}}, {{1, 1, 1, 0, 0}, {2, 0, 1, 0, 0}}},
  {{{3, 1, 1, 0, 0}, {0, 0, 1, 0, 0}}, {{0, 0, 1, 0, 0}, {3, 3, 1, 0, 0}}},
  {{{1, 1, 1, 0, 0}, {2, 0, 1, 0, 0}}, {{1, 1, 1, 0, 0}, {2, 0, 1, 0, 0}}},
  {{{1, 1, 1, 0, 0}, {2, 0, 1, 0, 0}}, {{0, 0, 1, 0, 0}, {3, 3, 1, 0, 0}}},
  {{{0, 0, 1, 0, 0}, {3, 3, 1, 0, 0}}, {{0, 0, 1, 0, 0}, {3, 3, 1, 0, 0}}},
  {{{2, 0, 1, 0, 0}, {1, 1, 1, 2, 2}}, {{2, 0, 1, 0, 0}, {1, 1, 1, 2, 2}}},
  {{{2, 0, 1, 0, 0}, {1, 1, 1, 2, 2}}, {{1, 1, 1, 0, 0}, {2, 2, 1, 2, 2}}},
  {{{2, 0, 1, 0, 0}, {1, 1, 1, 2, 2}}, {{0, 0, 1, 0, 0}, {3, 1, 1, 2, 2}}},
  {{{1, 1, 1, 0, 0}, {2, 2, 1, 2, 2}}, {{1, 1, 1, 0, 0}, {2, 2, 1, 2, 2}}},
  {{{1, 1, 1, 0, 0}, {2, 2, 1, 2, 2}}, {{0, 0, 1, 0, 0}, {3, 1, 1, 2, 2}}},
  {{{0, 0, 1, 0, 0}, {3, 1, 1, 2, 2}}, {{0, 0, 1, 0, 0}, {3, 1, 1, 2, 2}}},
  {{{0, 0, 1, 0, 0}, {3, 3, 1, 3, 3}}, {{0, 0, 1, 0, 0}, {3, 3, 1, 3, 3}}},
  {{{1, 1, 1, 0, 0}, {2, 2, 1, 4, 4}}, {{1, 1, 1, 0, 0}, {2, 2, 1, 4, 4}}},
  {{{1, 1, 1, 0, 0}, {2, 2, 1, 4, 4}}, {{0, 0, 1, 0, 0}, {3, 3, 1, 4, 4}}},
  {{{0, 0, 1, 0, 0}, {3, 3, 1, 4, 4}}, {{0, 0, 1, 0, 0}, {3, 3, 1, 4, 4}}},
  {{{0, 0, 1, 0, 0}, {3, 3, 1, 6, 6}}, {{0, 0, 1, 0, 0}, {3, 3, 1, 6, 6}}}}
```

```
In[7]:= genintsHb[{0, 2}, 3]
```

```
Out[7]:= {v[sss,sss;0], v[sss,sdd;0], v[sss,ddd;0], v[sdd,sdd;0], v[sdd,ddd;0],
  v[ddd,ddd;0], v[ssd,ssd;2], v[ssd,sdd;2], v[ssd,ddd;2], v[sdd,sdd;2], v[sdd,ddd;2],
  v[ddd,ddd;2], v[ddd,ddd;3], v[sdd,sdd;4], v[sdd,ddd;4], v[ddd,ddd;4], v[ddd,ddd;6]}
```

FIG. 2: With `genintHb[{0,2},3]` one generates the 17 Hermitian three-boson interaction matrix elements in *sd*-IBM, which are listed in the notation that is used internally in `ibm.m`. The same 17 interactions are generated with `genintsHb[{0,2},3]` in a more compact notation.

4. Definition of the Hamiltonian

There exist three ways to express a Hamiltonian in $\ell_1 \dots \ell_p$ -IBM, either in terms of operators in normal-ordered form, with boson creation operators on the left and annihilation operators on the right, or in multipole form, as products of bilinear operators of the form $b_\ell^\dagger b_{\ell'}$, or as a combination of both.

4.1 Hamiltonian in normal order

The statement `hamb[{ $\ell_1, \dots, \ell_p$ },  $k, N, J]$` returns the matrix elements of the Hamiltonian between N -boson states coupled to angular momentum J in terms of all possible k -body

```

In[3]:= << Documents/mathematica/ibm_sd.m
In[4]:= dimb[{0, 2}, 4, 0]
Out[4]= 4

In[5]:= hamb[{0, 2}, 1, 4, 0] + hamb[{0, 2}, 2, 4, 0] + hamb[{0, 2}, 3, 4, 0] + hamb[{0, 2}, 4, 4, 0]
Out[5]= {4 e[s] + 6 v[ss, ss; 0] + 4 v[sss, sss; 0] + v[ssss, ssss; 0],
  sqrt(6) v[ss, dd; 0] + 2 sqrt(2) v[sss, sdd; 0] + v[ssss, ssdd; 0],
  2 v[sss, ddd; 0] + v[ssss, sddd; 0], v[ssss, dddd; 0], 2 e[d] + 2 e[s] + v[dd, dd; 0] +
  2 v[sdd, sdd; 0] + 4 v[sd, sd; 2] + v[ssdd, ssdd; 0] + 2 v[ssd, ssd; 2] + v[ss, ss; 0],
  2 sqrt(3) v[sd, dd; 2] + sqrt(2) v[sdd, ddd; 0] + v[ssdd, sddd; 0] + sqrt(6) v[ssd, sdd; 2],
  sqrt(14/5) v[ss, dd; 0] + 2 sqrt(2) v[ssd, ddd; 2] + v[ssdd, dddd; 0],
  3 e[d] + e[s] + 3 v[dd, dd; 2] + v[ddd, ddd; 0] + v[sddd, sddd; 0] + 3 v[sdd, sdd; 2] + 3 v[sd, sd; 2],
  2 sqrt(6/7) v[sd, dd; 2] + 2 sqrt(3) v[sdd, ddd; 2] + v[sddd, dddd; 0],
  4 e[d] + (14 v[dd, dd; 0])/5 + (8 v[dd, dd; 2])/7 + (72 v[dd, dd; 4])/35 + 4 v[ddd, ddd; 2] + v[dddd, dddd; 0]}

```

FIG. 3: The number of states for $N = 4$ sd bosons with angular momentum $J = 0$ is four. With $\sum_{k=1}^4 \text{hamb}[\{0, 2\}, k, 4, 0]$ one calculates an array of length 10, which contains the single-boson energies and the two-, three- and four-boson interaction matrix elements.

interaction matrix elements, which are displayed in the spectroscopic notation explained in section 3. The output is an array of length $\frac{1}{2}d(d+1)$, where d is the matrix dimension, corresponding to the sequence $\{h_{11}, \dots, h_{1d}, h_{22}, \dots, h_{2d}, \dots, h_{dd}\}$. A $d \times d$ square matrix is constructed with `hamsqb` $[\{\ell_1, \dots, \ell_p\}, k, N, J]$. In general, the Hamiltonian matrix generated by `hamsqb` cannot be diagonalised since its elements are expressed symbolically in terms of the k -body interaction matrix elements.

The statement `hovb` $[\{\ell_1, \dots, \ell_p\}, N, J, \text{int}]$ returns the matrix elements of a single interaction ‘int’ between N -boson states coupled to angular momentum J . The ‘int’ is an element of the array generated by `genintb` (not `genintsb`). For example, if `int` is `genintsb` $[\{\ell_1, \dots, \ell_p\}, 3][[1]]$ then `hovb` $[\{\ell_1, \dots, \ell_p\}, N, J, \text{int}]$ calculates all matrix elements of the first three-body interaction between N -boson states. The output of `hovb` is an array of length $\frac{1}{2}d(d+1)$ with elements that are the square root of a rational number.

```

In[3]:= << Documents/mathematica/ibm_sd.m

In[7]:= genintsb[{0, 2}, 2]

Out[7]:= {v[ss,ss;0], v[ss,dd;0], v[dd,ss;0], v[dd,dd;0],
          v[sd,sd;2], v[sd,dd;2], v[dd,sd;2], v[dd,dd;2], v[dd,dd;4]}

In[10]:= dimb[{0, 2}, 10, 0]

Out[10]:= 14

In[15]:= (hovb[{0, 2}, 10, 0, genintb[{0, 2}, 2][[1]] +
          Sqrt[5] * hovb[{0, 2}, 10, 0, genintb[{0, 2}, 2][[2]]] +
          Sqrt[5] * hovb[{0, 2}, 10, 0, genintb[{0, 2}, 2][[3]]] +
          5 * hovb[{0, 2}, 10, 0, genintb[{0, 2}, 2][[4]]]) / 2

Out[15]:= {45/2, 15/2, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 33/2, 0, 7*sqrt(2), 0, 0, 0, 0, 0, 0, 0, 0, 0, 21/2, 0,
          sqrt(231)/2, 0, 0, 0, 0, 0, 0, 0, 0, 29/2, 0, 9*sqrt(5)/2, 0, 0, 0, 0, 0, 0, 0, 21/2, 0, 0, sqrt(65),
          0, 0, 0, 0, 0, 0, 33/2, 0, 0, sqrt(66), 0, 0, 0, 0, 0, 0, 3, 0, 0, sqrt(51)/2, 0, 0, 0, 0, 29/2, 0, 0,
          3*sqrt(15)/2, 0, 0, 0, 45/2, 0, 0, 0, sqrt(65)/2, 0, 9, 0, 0, 0, sqrt(19)/2, 45/2, 0, 0, 0, 0, 0, 0, 65/2, 0, 19}

In[16]:= eigenvalues[sqh[%]]

Out[16]:= {0, 0, 0, 0, 11, 11, 11, 20, 20, 20, 27, 27, 32, 35}

```

FIG. 4: With `genintsb[{0,2},2]` one generates all two-boson interaction matrix elements in *sd*-IBM. The combination $\frac{1}{2}(v[ss,ss;0] + \sqrt{5}(v[ss,dd;0] + v[dd,ss;0]) + 5v[dd,dd;0])$ corresponds to the *s* + *d* pairing interaction. Diagonalisation of the Hamiltonian matrix for $N = 10$ bosons yields the SO(6) eigenvalues $(N - \sigma)(N + \sigma + 4)$ with $\sigma = 10, 8, 6, 4, 2$ and 0.

4.2 Multipole Hamiltonian

The statement `hopb[{ $\ell_1, \dots, \ell_p$ },  $N, J, \text{hop}$ ]` returns the matrix elements of the scalar operator ‘hop’ between N -boson states coupled to angular momentum J . This operator is a combination of k bilinear operators $(b_\ell^\dagger \times \tilde{b}_{\ell'})^{(\lambda)}$, coupled in angular momentum according to the convention of [1] and should be predefined either in `ibm.m`, if it can be defined generically for all IBM versions, and/or in a file for a particular realisation of $\ell_1 \dots \ell_p$ -IBM.

Here is the list of generic scalar operators defined in `ibm.m` for $k = 1$ and $k = 2$:

- `obn1`: $\hat{n}_{\ell_1}$, counting the number of ℓ_1 bosons;
- `obn2`: $\hat{n}_{\ell_2}$, counting the number of ℓ_2 bosons;

```

In[3]:= << Documents/mathematica/ibm_sd.m
In[35]:= acb[{0, 2}, obQ, 2, 2, 2]
Out[35]= -  $\frac{\sqrt{7}}{2}$ 

In[36]:= 2 * hopb[{0, 2}, 8, 0, obQQ]
Out[36]= {80, 8  $\sqrt{35}$ , 0, 0, 0, 0, 0, 0, 0, 0, 126, -12  $\sqrt{14}$ , 4  $\sqrt{210}$ , 0, 0, 0, 0, 0, 0,
2  $\sqrt{5}$   $\left(\frac{21}{\sqrt{5}} + 8\sqrt{5}\right)$ , -8  $\sqrt{15}$ , 4  $\sqrt{110}$ , 0, 0, 0, 0, 0, 2  $\sqrt{5}$   $\left(\frac{54}{\sqrt{5}} + 4\sqrt{5}\right)$ , -8  $\sqrt{66}$ ,
36  $\sqrt{2}$ , 0, 0, 0, 0, 132, -8  $\sqrt{33}$ , -8  $\sqrt{39}$ , 4  $\sqrt{78}$ , 0, 0, 146, 0, -4  $\sqrt{286}$ , 8  $\sqrt{11}$ ,
0, 92, -16  $\sqrt{2}$ , 0, 4  $\sqrt{17}$ , 2  $\sqrt{5}$   $\left(\frac{29}{\sqrt{5}} + 6\sqrt{5}\right)$ , -8  $\sqrt{26}$ , -8  $\sqrt{34}$ , 120, 0, 66}

In[37]:= eigenvalues[sqh[%]]
Out[37]= {10, 28, 46, 76, 88, 106, 130, 148, 214, 304}

In[38]:= e2su3[{16, 0}]
Out[38]= 304

```

FIG. 5: First check whether the coefficient $\text{acb}[\{0, 2\}, \text{obQ}, 2, 2, 2]$ is set to its $\text{SU}(3)$ value. With $\text{hopb}[\{0, 2\}, 8, 0, \text{obQQ}]$ one calculates the matrix elements of $\hat{Q} \cdot \hat{Q}$ with eigenvalues appropriate for the $\text{SU}(3)$ limit of sd -IBM for boson number $N = 8$ and angular momentum $J = 0$.

- obn3: $\hat{n}_{\ell_3}$, counting the number of ℓ_3 bosons;
- obn4: $\hat{n}_{\ell_4}$, counting the number of ℓ_4 bosons;
- obn1n1: $\hat{n}_{\ell_1} \hat{n}_{\ell_1}$;
- obn1n2: $\hat{n}_{\ell_1} \hat{n}_{\ell_2}$;
- obn2n2: $\hat{n}_{\ell_2} \hat{n}_{\ell_2}$;
- obLL: $\hat{L}^2$ where $\hat{L}_\mu$ are the three angular momentum operators;
- obQQ: $\hat{Q}^2$ where $\hat{Q}_\mu$ are the five quadrupole operators;
- ob00: $\hat{T}_3^2$ where $\hat{T}_{3\mu}$ are the seven octupole operators;
- obHH: $\hat{T}_4^2$ where $\hat{T}_{4\mu}$ are the nine hexadecapole operators.

More operators that are combination of $\hat{L}_\mu$ and $\hat{Q}_\mu$ with $k > 2$ are defined in `ibm.m` and are listed in section 10.2.

Matrix elements of $\hat{n}_i$, $\hat{n}_i\hat{n}_j$ and $\hat{L}^2$ can be calculated without giving their explicit expression in terms boson creation and annihilation operators. This, in general, is not possible for other operators, which depend on a particular realisation of $\ell_1 \dots \ell_p$ -IBM. The full definition of the operators $\hat{Q}^2$, $\hat{T}_3^2$ and $\hat{T}_4^2$ must therefore be given in the file for a particular realisation of $\ell_1 \dots \ell_p$ -IBM. For example, in *sd*-IBM the octupole and hexadecapole operators are defined as $\text{obO} = \hat{T}_{3\mu} \equiv (d^\dagger \times \tilde{d})_\mu^{(3)}$ and $\text{obH} = \hat{T}_{4\mu} \equiv (d^\dagger \times \tilde{d})_\mu^{(4)}$, which is obtained with the following statements in the file `ibm_sd.m`:

$$\text{acb}[\{0, 2\}, \text{obO}, 3, 2, 2] = 1, \text{acb}[\{0, 2\}, \text{obH}, 4, 2, 2] = 1. \quad (1)$$

The quadrupole operator in *sd*-IBM, on the other hand, has three parts. Its standard definition,

$$\text{obQ} = (s^\dagger \times \tilde{d} + d^\dagger \times \tilde{s})_\mu^{(2)} + \chi(d^\dagger \times \tilde{d})_\mu^{(2)}, \quad (2)$$

is obtained with

$$\text{acb}[\{0, 2\}, \text{obQ}, 2, 1, 2] = 1, \text{acb}[\{0, 2\}, \text{obQ}, 2, 2, 1] = 1, \text{acb}[\{0, 2\}, \text{obQ}, 2, 2, 2] = \chi, \quad (3)$$

where χ can be a symbol or a number. The argument λ in $\text{acb}[\{\ell_1, \dots, \ell_p\}, \text{hop}, \lambda, i, j]$ refers to the angular momentum of the operator $(b_\ell^\dagger \times \tilde{b}_{\ell'})^{(\lambda)}$ while i and j define the constituent bosons, that is, $\ell = \ell_i$ and $\ell' = \ell_j$.

The definition of multipole operators therefore proceeds in two steps. In a first, the *generic* structure of the operator is defined in `ibm.m`. It involves a number of bilinear operators such as `obL`, `obQ`, `obO` and `obH`, which are coupled according to a particular sequence of angular momenta. In a second step, the *specific* structure of the operators `obL`, `obQ`, `obO` and `obH`, that is, their coefficients ‘`acb`’, is defined in the file for a particular realisation of $\ell_1 \dots \ell_p$ -IBM, such as `ibm_sd.m` or `ibm_sdg.m`. It is possible to introduce several operators with the same angular momentum λ . For example, in *sdg*-IBM there exist several limits with different quadrupole operators; they are distinguished by introducing `obQa`, `obQb`, etc.

Additional multipole operators (mostly Casimir operators) that are specific for particular realisations of $\ell_1 \dots \ell_p$ -IBM are listed in section 8.

4.3 From a multipole to a normal-ordered Hamiltonian

As explained in [1], the algorithm to calculate matrix elements of a normal-ordered interaction is more efficient than that to obtain them from a multipole Hamiltonian. It is there-


```

In[43]:= << Documents/mathematica/ibm_sd.m
In[44]:= acb[{0, 2}, obQ, 2, 2, 2]
Out[44]= chi

In[45]:= multonorb[{0, 2}, {{a, obQQ}, {b, obLL}}]
Out[45]= {E0 -> 0, e[s] -> 5 a, e[d] -> 6 b + a (1 + chi^2), v[ss,ss;0] -> 0, v[ss,dd;0] -> 2 sqrt(5) a,
          v[dd,ss;0] -> 2 sqrt(5) a, v[dd,dd;0] -> -12 b + 2 a chi^2, v[sd,sd;2] -> 2 a, v[sd,dd;2] -> 2 sqrt(2) a chi,
          v[dd,sd;2] -> 2 sqrt(2) a chi, v[dd,dd;2] -> -6 b - (3 a chi^2)/7, v[dd,dd;4] -> 8 b + (4 a chi^2)/7}

```

FIG. 6: The coefficient $\text{acb}[\{0, 2\}, \text{obQ}, 2, 2, 2]$ was redefined and has been set to a symbolic value χ in the file `ibm_sd.m`. With `multonorb[\{0, 2\}, \{\{a, \text{obQQ}\}, \{b, \text{obLL}\}\}` the multipole Hamiltonian $a \hat{Q} \cdot \hat{Q} + b \hat{L} \cdot \hat{L}$ in *sd*-IBM is converted to normal order.

fore of interest to convert the multipole Hamiltonian from the start to its normal-ordered representation. In `ibm.m` the multipole Hamiltonian is specified by an array called ‘hops’ with elements of the form $\{c_{\text{hop}}, \text{hop}\}$, where c_{hop} is the (numerical or symbolic) coefficient that multiplies the operator ‘hop’. The statement `multonorb[\{\ell_1, \dots, \ell_p\}, \text{hops}]` returns a substitution rule for normal-ordered interactions in terms of the coefficients c_{hop} .

4.4 A combined normal-ordered and multipole Hamiltonian

With `hopvb[\{\ell_1, \dots, \ell_p\}, N, J, \text{hops}, \text{ints}]` it is possible to construct the matrix of a Hamiltonian that is an arbitrary linear combination of operators in multipole and in normal-ordered form. The array ‘hops’ has the same structure as in section 4.3 and, similarly, ‘ints’ consists of elements of the form $\{c_{\text{int}}, \text{int}\}$, where c_{int} is the coefficient that multiplies the interaction ‘int’. There exist two versions of `hopvb`. The first, `hopvb1`, converts the multipole part of the Hamiltonian to normal order while the second, `hopvb2`, does not. The default `hopvb` is set to `hopvb1` since it is the more efficient algorithm.

5. Calculation of energy spectra

Energy spectra are obtained with `specb[\{\ell_1, \dots, \ell_p\}, N, \{J_1, J_2, \dots\}, n, \text{hops}, \text{ints}, c]`, where N is the number of bosons, $\{J_1, J_2, \dots\}$ defines the sequence of angular momenta to be calculated, n is the number of eigenstates per angular momentum and c controls the

```

In[16]:= << Documents/mathematica/ibm_sd.m

In[17]:= acb[{0, 2}, obQ, 2, 2, 2]

Out[17]= -0.5

In[19]:= specb[{0, 2}, 10, {0, 2, 3, 4}, 5, {{0.25, obn2}, {-0.05, obQQ}, {0.05, obLL}}, {}, 2]

Ground-state energy: -6.81774

J=0 (dim=14) Energies: {0., 1.8985, 2.72166, 3.59959, 4.31793}
Eigenvector yrast state: {-0.143496, -0.413053, 0.276265, -0.567241, 0.382184, -0.414897,
-0.0771723, 0.238635, -0.15175, -0.0475035, 0.0623691, 0.00366388, -0.0187977, -0.00800251}

J=2 (dim=22) Energies: {0.449284, 1.7198, 2.50835, 3.19199, 3.47071}
Eigenvector yrast state: {-0.214798, 0.214588, -0.472777, 0.373823, 0.185973, -0.50077, -0.108469,
0.30717, 0.201992, -0.272816, -0.0908055, -0.0297174, 0.120732, 0.0917645, 0.0100221,
-0.0653844, -0.0277458, -0.0123303, 0.0157195, 0.0131126, 0.00230925, 0.000589947}

J=3 (dim=8) Energies: {2.2545, 4.20084, 5.17463, 6.22034, 7.28567}
Eigenvector yrast state:
{-0.557711, -0.690264, 0.199445, -0.388585, 0.113118, -0.0921922, -0.0105713, 0.0176206}

J=4 (dim=25) Energies: {1.48094, 2.84601, 3.64636, 3.75908, 4.42845}
Eigenvector yrast state:
{-0.303199, 0.312227, -0.523244, 0.098219, 0.430302, 0.129602, -0.426711, 0.107388, -0.0891674,
0.26791, 0.107604, -0.0181617, -0.16669, 0.0490802, -0.0547364, -0.0112716, 0.0698669, 0.0326425,
-0.00757878, 0.00426036, -0.0215944, 0.0070515, -0.00920027, -0.00257869, 0.000382699}

```

FIG. 7: The coefficient $\text{acb}[\{0, 2\}, \text{obQ}, 2, 2, 2]$ was redefined and has been set to -0.5 in `ibm_sd.m`. With `specb[\{0, 2\}, 10, \{0, 2, 3, 4\}, 5, \{\{0.25, \text{obn2}\}, \{-0.05, \text{obQQ}\}, \{0.05, \text{obLL}\}\}, \{\}, 2]` the spectrum of the Hamiltonian $\epsilon_d \hat{n}_d + \kappa_2 \hat{Q} \cdot \hat{Q} + \kappa_1 \hat{L} \cdot \hat{L}$ is calculated for $N = 10$ bosons and the angular momenta $J = 0, 2, 3$ and 4 , with the parameters $\epsilon_d = 0.25$, $\kappa_2 = -0.05$ and $\kappa_1 = 0.05$.

output. The latter parameter can have the following values:

- $c = 1$: Prints ground-state energy and excitation energies of n eigenstates for each J .
- $c = 2$: Prints in addition the eigenvector of the yrast eigenstate for each J .
- $c = 3$: Prints in addition the eigenvectors of the n lowest eigenstates for each J .
- $c = 4$: As for $c = 1$ and outputs n eigenvalues and eigenvectors.

6. Calculation of electromagnetic transition rates and moments

The statement `topb[\{\ell_1, \dots, \ell_p\}, N_i, J_i, N_f, J_f, top]` calculates reduced matrix elements of a generic tensor operator ‘top’, which is predefined in `ibm.m` and/or in a file for a particular realisation of $\ell_1 \dots \ell_p$ -IBM. The operator does not necessarily conserve boson number, which

```

In[26]:= << Documents/mathematica/ibm_sd.m
In[21]:= acb[{0, 2}, obQ, 2, 2, 2]
Out[21]= 0

In[27]:= emtrb[{0, 2}, 8, 2, 3, 0, 2, {{-0.1, obQQ}, {0.05, ob00}, {0.02, obLL}}, {}, obE2, 2]
J=2 (dim=15) Energies: {-9.15, -8.7, -7.35}
Ground-state energy: -9.6
J=0 (dim=10) Energies: {-9.6, -8.25}
B[obE2;2(1)->0(1)]=19.2 eb2
B[obE2;2(2)->0(1)]=0
B[obE2;2(3)->0(1)]=0
B[obE2;2(1)->0(2)]=0
B[obE2;2(2)->0(2)]=5.6 eb2
B[obE2;2(3)->0(2)]=10. eb2

```

FIG. 8: The quadrupole operator `obQ` was redefined by setting `acb[{0,2},obQ,2,2,2] = 0` in `ibm_sd.m`. With `emtrb[{0,2},8,2,3,0,2,{{-0.1,obQQ},{0.05,ob00},{0.02,obLL}},{},obE2,2]` first the energies of three $J = 2$ and two $J = 0$ states are calculated for $N = 8$ bosons with the Hamiltonian $\kappa_2 \hat{Q} \cdot \hat{Q} + \kappa_3 \hat{T}_3 \cdot \hat{T}_3 + \kappa_1 \hat{L} \cdot \hat{L}$ with the parameters $\kappa_2 = -0.1$, $\kappa_3 = -0.05$ and $\kappa_1 = 0.02$. Subsequently, the $B(E2; 2^+ \rightarrow 0^+)$ values are calculated with the quadrupole operator $\text{obE2} = e_b(s^\dagger \times \tilde{d} + d^\dagger \times \tilde{s})_\mu^{(2)}$, as defined in `ibm_sd.m`.

therefore must be specified for the initial and the final states along with their angular momenta J_i and J_f . The routine `topb` returns a matrix between initial and final states generated by `basb[{ $\ell_1, \dots, \ell_p$ },  $N_i, J_i]$  and basb[{ $\ell_1, \dots, \ell_p$ },  $N_f, J_f]$ , respectively.`

Electromagnetic transitions between eigenstates of a boson Hamiltonian are calculated with `emtrb[{ $\ell_1, \dots, \ell_p$ },  $N, J_i, n_i, J_f, n_f, \text{hops}, \text{ints}, \text{top}, c]$` . Since the boson number does not change between the initial and final states, it can be declared only once as N . The Hamiltonian is defined in ‘hops’ and ‘ints’, according to the convention explained in section 4.4, and with this Hamiltonian n_i (n_f) eigenstates with angular momentum J_i (J_f) are calculated. Subsequently, the matrix elements of the boson-number conserving operator ‘top’ between the initial and final eigenstates are evaluated. The parameter c controls the output with the following options:

- $c = 1$: Prints reduced matrix elements between initial and final states.
- $c = 2$: Prints $B(E2)$ values for transitions from initial to final states.

- $c = 3$: Outputs reduced matrix elements between initial and final states.

Electromagnetic moments of eigenstates of a boson Hamiltonian are calculated with `emmob` $[\{\ell_1, \dots, \ell_p\}, N, J, n, \text{hops}, \text{ints}, \text{top}, c]$, which works along the same conventions but now with the same state in bra and ket. The parameter c controls the output with the following options:

- $c = 1$: Prints reduced matrix elements of n eigenstates with angular momentum J .
- $c = 2$: Prints electromagnetic moments of n eigenstates with angular momentum J .
- $c = 3$: Outputs reduced matrix elements of n eigenstates with angular momentum J .

7. Parity

If some of the bosons have negative parity, N -bosons states with different parity may occur. Conservation of parity implies that the Hamiltonian matrix is separated in two unconnected blocks that can be treated separately. The parity of the bosons can be declared with $\{p_1, \dots, p_p\}$, where $p_i = \pm 1$ refers to the parity of the boson with angular momentum ℓ_i . As a general rule, conservation of parity is consistently imposed by inserting the parity of the bosons $\{p_1, \dots, p_p\}$ after the argument containing their angular momenta $\{\ell_1, \dots, \ell_p\}$ and by specifying the parity P after the angular momentum J . For example, as discussed in section 2, the statement `basb` $[\{\ell_1, \dots, \ell_p\}, N, J]$ constructs a basis for N bosons coupled to angular momentum J . To impose parity conservation, it should be replaced by `basb` $[\{\ell_1, \dots, \ell_p\}, \{p_1, \dots, p_p\}, N, J, P]$.

8. Particular realisations of $\ell_1 \dots \ell_p$ -IBM

8.1 *sd*-IBM

The energy spectrum of a Hamiltonian in *sd*-IBM can be obtained with the statement `specsd` $[N, \{J_1, J_2, \dots\}, n, \text{hops}, \chi, c]$, where N is the number of *s* and *d* bosons, $\{J_1, J_2, \dots\}$ defines the sequence of angular momenta to be calculated, n is the number of eigenstates per angular momentum and c controls the output, with options as defined in section 5. The argument ‘hops’ is an array with elements of the form $\{c_{\text{hop}}, \text{hop}\}$, where c_{hop} is a numerical

coefficient that multiplies the operator ‘hop’. All standard interaction operators of *sd*-IBM have been predefined. With the input

$$\begin{aligned} \text{hops} = \{ \{ \epsilon_s, \text{obn1} \}, \{ \epsilon_d, \text{obn2} \}, \{ \lambda_s, \text{obn1n1} \}, \{ \lambda_d, \text{obn2n2} \}, \\ \{ \kappa_0, \text{obPPsd} \}, \{ \kappa_1, \text{obLL} \}, \{ \kappa_2, \text{obQQ} \}, \{ \kappa_3, \text{ob00} \}, \{ \kappa_4, \text{obHH} \} \}, \end{aligned} \quad (4)$$

the spectrum of the Hamiltonian

$$\begin{aligned} \hat{H}_{sd} = \epsilon_s \hat{n}_s + \epsilon_d \hat{n}_d + \lambda_s \hat{n}_s^2 + \lambda_d \hat{n}_d^2 \\ + \kappa_0 \hat{P}_+ \hat{P}_- + \kappa_1 \hat{L} \cdot \hat{L} + \kappa_2 \hat{Q}^\chi \cdot \hat{Q}^\chi + \kappa_3 \hat{T}_3 \cdot \hat{T}_3 + \kappa_4 \hat{T}_4 \cdot \hat{T}_4, \end{aligned} \quad (5)$$

is calculated in terms of the following operators:

- the *s*-boson number operator $\hat{n}_s = s^\dagger \tilde{s}$;
- the *d*-boson number operator $\hat{n}_d = d^\dagger \cdot \tilde{d} = \sqrt{5}(d^\dagger \times \tilde{d})_0^{(0)}$;
- the *sd*-pairing operators $\hat{P}_+ = \frac{1}{2}(s^\dagger s^\dagger - d^\dagger \cdot d^\dagger)$ and $\hat{P}_- = \hat{P}_+^\dagger$;
- the angular momentum operators $\hat{L}_\mu = \sqrt{10}(d^\dagger \times \tilde{d})_\mu^{(1)}$;
- the quadrupole operators $\hat{Q}_\mu^\chi = (s^\dagger \times \tilde{d} + d^\dagger \times \tilde{s})_\mu^{(2)} + \chi(d^\dagger \times \tilde{d})_\mu^{(2)}$;
- the octupole operators $\hat{T}_{3\mu} = (d^\dagger \times \tilde{d})_\mu^{(3)}$;
- the hexadecapole operators $\hat{T}_{4\mu} = (d^\dagger \times \tilde{d})_\mu^{(4)}$.

If a term in the Hamiltonian (5) is zero, it can be omitted from the list ‘hops’. Although the Hamiltonian matrix can be constructed for symbolic values of the parameters, the diagonalisation of this matrix can only be carried if *all* parameters $\{\epsilon_\ell, \lambda_\ell, \kappa_i\}$ have numerical values. The numerical value of the parameter χ in the quadrupole operator must be declared separately as the fifth argument in **specsd**. The multipole operators in the Hamiltonian (5) must be defined with the following statements:

$$\begin{aligned} \text{acb}[\{0, 2\}, \text{obL}, 1, 2, 2] &= \text{Sqrt}[10], \\ \text{acb}[\{0, 2\}, \text{obQ}, 2, 1, 2] &= 1, \text{acb}[\{0, 2\}, \text{obQ}, 2, 2, 1] = 1, \text{acb}[\{0, 2\}, \text{obQ}, 2, 2, 2] = \text{chi}, \\ \text{acb}[\{0, 2\}, \text{ob0}, 3, 2, 2] &= 1, \\ \text{acb}[\{0, 2\}, \text{obH}, 4, 2, 2] &= 1. \end{aligned} \quad (6)$$

These are the default definitions in the file `ibm_sd.m` and they should not be modified to ensure the proper functioning of `specs`.

Electromagnetic transitions between eigenstates of a Hamiltonian in *sd*-IBM are calculated with `emtrsd`[$N, J_i, n_i, J_f, n_f, \text{hops}, \chi_h, \text{top}, \chi_t, c$]. The Hamiltonian is defined with ‘hops’ and χ_h , as explained above, and with this Hamiltonian n_i (n_f) eigenstates with angular momentum J_i (J_f) are calculated. Subsequently, the matrix elements of the boson-number conserving operator ‘top’ between the initial and final eigenstates are evaluated. For *E2* transitions the operator `top=obE2` is further specified by giving a numerical value for χ_t , which may or may not coincide with χ_h . The parameter c controls the output as explained in section 6. Electromagnetic moments of eigenstates of a Hamiltonian in *sd*-IBM are calculated with `emmosd`[$N, J, n, \text{hops}, \chi_h, \text{top}, \chi_t, c$], which works along the same conventions but now with the same state in bra and ket. Presently, two electromagnetic-transition operators are defined with the statements

$$\text{acb}[\{0, 2\}, \text{obM1}, 1, 2, 2] = \text{Sqrt}[10] g_b, \quad (7)$$

$$\text{acb}[\{0, 2\}, \text{obE2}, 2, 1, 2] = e_b, \text{acb}[\{0, 2\}, \text{obE2}, 2, 2, 1] = e_b, \text{acb}[\{0, 2\}, \text{obE2}, 2, 2, 2] = e_b \chi,$$

which should not be modified to ensure the proper functioning of `emtrsd` and `emmosd`.

A number of additional operators specific to *sd*-IBM are defined in `IBM_sd.m`. They include the linear Casimir operators `c1u6sd` and `c1u5sd`, which are nothing but the number operators $\hat{N}$ and $\hat{n}_d$. Also defined are the quadratic operators `c2u6sd`, `c2u5sd` and `c1u6u5sd`, to be identified with $\hat{N}(\hat{N} + 5)$, $\hat{n}_d(\hat{n}_d + 4)$, and $\hat{N}\hat{n}_d$, respectively, and four more quadratic Casimir operators, namely `c2su3sd`, `c2so6sd`, `c2so5sd` and `c2so3sd`. The quadratic Casimir operator of $\text{SO}(6)$ is related to the pairing operator of *sd*-IBM by `c2so6sd =  $\hat{N}(\hat{N} + 4) - 4 \text{obPPsd}$` .

The 17 normal-ordered, normalised three-boson interactions of *sd*-IBM are defined with the notation `ob $\ell_1\ell_2\ell_3L\ell'_1\ell'_2\ell'_3L$` , where ℓ_i (ℓ'_i) refers to the angular momenta of the three bosons in the bra (ket) and L to their total angular momentum. Thus, for example, the normalised interaction between three *d* bosons coupled to $L = 3$ is denoted as `obddd3ddd3`. Other higher-order operators include the cubic Casimir operator of $\text{SU}(3)$, `c3su3sd`, and the square of the quadratic Casimir operator of $\text{SU}(3)$, `c22su3sd`. Finally, `IBM_sd.m` defines three additional operators, namely `obP0nsP0sd`, `obP0ndP0sd` and `obG3G3sd`, following the conventions of Leviatan and Shapira [2].

8.2 *sdg*-IBM

A number of additional operators specific to *sdg*-IBM are defined in `IBM_sdg.m`. They are the Casimir operators of two limits, $SU(3)$ and $SU(5)$, which arise as subalgebras of the dynamical algebra $U(15)$ of *sdg*-IBM. Two quadratic Casimir operators, `c2su3sdg` and `c2so3sdg`, are defined in the $SU(3)$ limit while the $SU(5)$ limit results from a combination of three Casimir operators, `c2su5sdg`, `c2so5sdg` and `c2so3sdg`, the latter being common to both limits.

9. Tests

9.1 *Coefficients of fractional parentage*

A test of the CFPs for a boson with angular momentum ℓ is provided by running `testcfps` $[\ell, n, m]$, which generates all CFPs $[\ell^k(v_k\alpha_k J_k), \ell^m(v_m\alpha_m J_m) | \ell^n v_n \alpha_n J_n]$, where $k = n - m$, and checks their orthonormality. If successful, `testcfps` $[\ell, n, m]$ prints the values of J_n , for which it has finished the check. If not, it prints a warning. Be aware that the number of CFPs can be very large as the values of ℓ , n and/or m increase.

9.2 *Multipole and normal-ordered Hamiltonians*

A good test is to compare the results of `hopvb1` $[\{\ell_1, \dots, \ell_p\}, N, J, \text{hops}, \text{ints}]$ with those of `hopvb2` $[\{\ell_1, \dots, \ell_p\}, N, J, \text{hops}, \text{ints}]$. The two methods provide entirely different recursive schemes to calculate matrix elements in $\ell_1 \dots \ell_p$ -IBM but should give identical Hamiltonian matrices.

9.3 *Symmetry limits*

The file `IBM_sd.m` contains the three routines `specu5sd` $[N, J]$, `specsu3sd` $[N, J]$ and `specso6sd` $[N, J]$. They calculate with group-theoretical methods the eigenvalues for N

```
In[16]:= Simplify[%]
```

bosons with angular momentum J in the three limits of sd -IBM with the Hamiltonians

$$\hat{H}[\text{SO}(6)] = a\hat{C}_2[\text{SO}(6)] + b\hat{C}_2[\text{SO}(5)] + c\hat{C}_2[\text{SO}(3)], \quad (8)$$

Hamiltonian matrix, which is achieved with the following statements

$$\text{sqh}[\text{hopvb}[\{0, 2\}, N, J, \{\{a, \text{c2so6sd}\}, \{b, \text{c2so5sd}\}, \{c, \text{c2so3sd}\}\}, \{\}\]], \quad (9)$$

symbolic parameters a , b and c .


```

In[8]:= << Documents/mathematica/ibm_sdg.m

In[28]:= Simplify[hopvb[{0, 2, 4}, 3, 0, {{a, c2su5sdg}, {b, c2so5sdg}}, {}]]

Out[28]:= {84 a, 4  $\sqrt{15}$  a, 0, 0, 12  $\sqrt{3}$  a, 0, 0,  $\frac{2}{7}$  (293 a + 36 b),  $-\frac{12\sqrt{15}}{7}$  a,  $\frac{96}{7}$  a,  $\frac{24}{7}\sqrt{5}$  (a - b),
 $\frac{12\sqrt{110}}{7}$  a, 0,  $\frac{6}{49}$  (793 a + 318 b),  $\frac{24}{49}\sqrt{15}$  (17 a + 8 b), 0,  $-\frac{6}{49}\sqrt{66}$  (10 a + b), 0,
 $\frac{2}{49}$  (2122 a + 747 b),  $\frac{16\sqrt{5}}{7}$  a,  $-\frac{4}{7}\sqrt{110}$  (a + 2 b),  $\frac{8}{49}\sqrt{2145}$  (3 a + b),  $\frac{2}{7}$  (309 a + 20 b),
 $\frac{40\sqrt{22}}{7}$  a,  $\frac{4\sqrt{429}}{7}$  a,  $\frac{34}{49}$  (117 a + 11 b),  $-\frac{20}{49}\sqrt{78}$  (2 a + 3 b),  $\frac{6}{49}$  (597 a + 220 b)}

In[29]:= eigenvalues[sqh[%]]

Out[29]:= {42 a, 82 a, 132 a, 2 (21 a + 8 b), 2 (41 a + 8 b), 6 (22 a + 9 b), 2 (41 a + 17 b)}

In[30]:= specs5sdg[3, 0]

Out[30]:= {132 a, 132 a + 54 b, 82 a, 82 a + 16 b, 82 a + 34 b, 42 a, 42 a + 16 b}

In[31]:= Sort[Simplify[%]] - Sort[Simplify[%]]

Out[31]:= {0, 0, 0, 0, 0, 0, 0}

```

FIG. 10: First the matrix elements in the SU(5) limit of *sdg*-IBM are calculated with `hopvb` for $N = 3$ bosons and angular momentum $J = 0$. The resulting matrix is diagonalised. The same spectrum is found analytically with `specs5sdg[3, 0]`.

The file `IBM_sdg.m` contains the two routines `specs3sdg[N, J]` and `specs5sd[N, J]`. They calculate with group-theoretical methods the eigenvalues for N bosons with angular momentum J in the two of the limits of *sdg*-IBM with the Hamiltonians

$$\begin{aligned}
\hat{H}[\text{SU}(3)] &= a \hat{C}_2[\text{SU}(3)] + c \hat{C}_2[\text{SO}(3)], \\
\hat{H}[\text{SU}(5)] &= a \hat{C}_2[\text{SU}(5)] + b \hat{C}_2[\text{SO}(5)] + c \hat{C}_2[\text{SO}(3)].
\end{aligned} \tag{10}$$

10. Further details

10.1 Notation of higher-order ($k > 2$) interactions with `genints`

The statement `genintsb[{ $\ell_1, \dots, \ell_p$ },  $k$ ]` generates all k -body interactions between the bosons in a simplified notation. States in bra and ket of the matrix element are normalised and defined according to the convention of [1]. Besides the angular momenta of the bosons $\ell_{i_1} \dots \ell_{i_k}$ in bra and ket, and the total angular momentum L , other quantum numbers are given only if needed. For example, three-body interactions in *sd*-IBM are uniquely specified

by $\ell_{i_1}\ell_{i_2}\ell_{i_3}$ and L , and therefore the notation $v[sss, sss; 0]$, $v[sss, sdd; 0]$, $v[sss, ddd; 0]$, etc suffices. In contrast, four d bosons are not uniquely specified by their angular momentum for $L = 2$ or $L = 4$, in which case there are two states, one with seniority $\nu = 2$ and a second with $\nu = 4$. In the interaction matrix elements they are denoted as $dddd(2)$ and $dddd(4)$, respectively. If the angular momentum L and the seniority ν are not sufficient to characterise the state completely, the multiplicity index is added in the brackets. For example, three g bosons coupled to angular momentum $L = 6$ and with seniority $\nu = 3$ have multiplicity $d_{\ell=4}(\nu = 3, L = 6) = 2$, leading to the notation $ggg(31)$ and $ggg(32)$.

10.2 Predefined higher-order ($k > 2$) scalar multipole operators

The following scalar multipole operators with $k = 3$ are defined in `ibm.m`:

- `obQQQ`: $(\hat{Q} \times \hat{Q} \times \hat{Q})^{(0)}$;
- `obLQQ`, `obQLQ`, `obQQL`: $(\hat{L} \times \hat{Q} \times \hat{Q})^{(0)}$, $(\hat{Q} \times \hat{L} \times \hat{Q})^{(0)}$, $(\hat{Q} \times \hat{Q} \times \hat{L})^{(0)}$;
- `obLLQ`, `obLQL`, `obQLL`: $(\hat{L} \times \hat{L} \times \hat{Q})^{(0)}$, $(\hat{L} \times \hat{Q} \times \hat{L})^{(0)}$, $(\hat{Q} \times \hat{L} \times \hat{L})^{(0)}$;
- `obLLL`: $(\hat{L} \times \hat{L} \times \hat{L})^{(0)}$.

The scalar multipole operators with $k = 4$ defined in `ibm.m` are:

- `obLL $\lambda$ LL`: $((\hat{L} \times \hat{L})^{(\lambda)} \times (\hat{L} \times \hat{L})^{(\lambda)})^{(0)}$ for $\lambda = 0, 1, 2$;
- `obQQ $\lambda$ QQ`: $((\hat{Q} \times \hat{Q})^{(\lambda)} \times (\hat{Q} \times \hat{Q})^{(\lambda)})^{(0)}$ for $\lambda = 0, 1, 2, 3, 4$;
- `obLQ $\lambda$ LQ`: $((\hat{L} \times \hat{Q})^{(\lambda)} \times (\hat{L} \times \hat{Q})^{(\lambda)})^{(0)}$ for $\lambda = 1, 2, 3$.

10.3 Predefined non-scalar multipole operators for $k = 2$

The following non-scalar multipole operators with $k = 2$ are defined in `ibm.m`:

- `obLL $\lambda$` : $(\hat{L} \times \hat{L})^{(\lambda)}$ for $\lambda = 1, 2$;
- `obLQ $\lambda$` : $(\hat{L} \times \hat{Q})^{(\lambda)}$ for $\lambda = 1, 2, 3$;
- `obQQ $\lambda$` : $(\hat{Q} \times \hat{Q})^{(\lambda)}$ for $\lambda = 1, 2, 3, 4$.

-
- [1] P. Van Isacker, *Boson models with interactions of arbitrary order*, Symmetry **18** (2026) 348.
- [2] A. Leviatan, D. Shapira, *Algebraic benchmark for prolate–oblate coexistence in nuclei*, Phys. Rev. C **93** (2016), 051302(R).